

Python PI
INSTITUTE

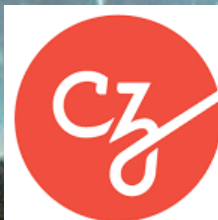


Napari

Tool for visualizing scientific images in 4D



UNIVERSIDAD
DE LA REPÚBLICA
URUGUAY

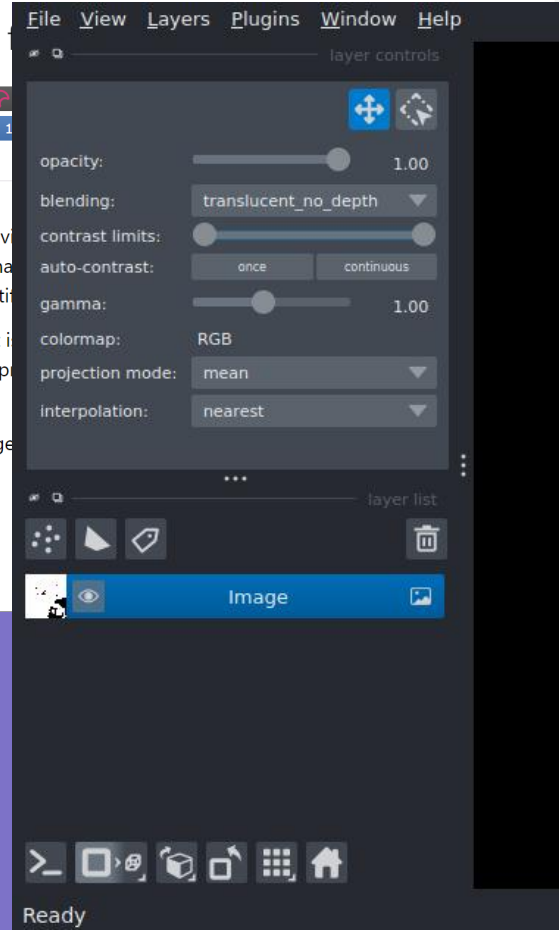
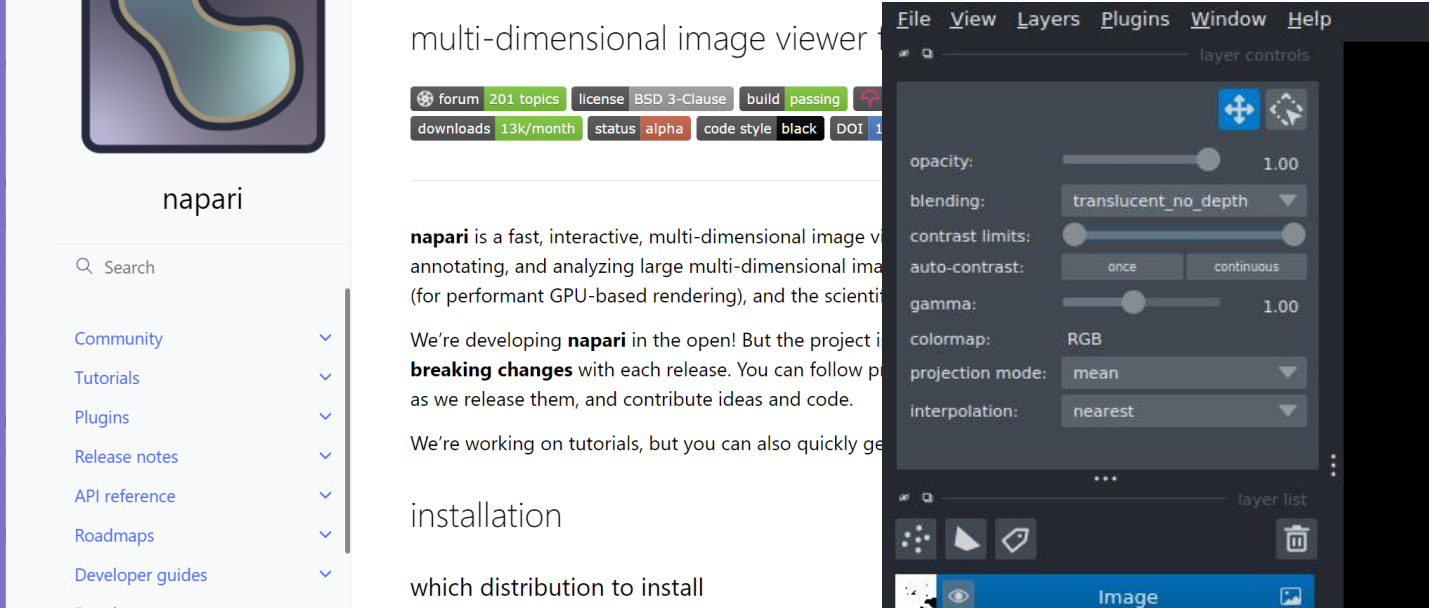
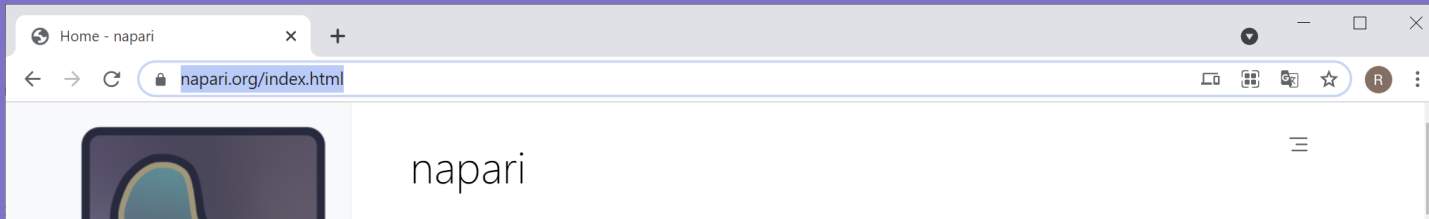


CHAN
ZUCKERBERG
INITIATIVE

Unidad de
Bioimagenología
Avanzada



napari community



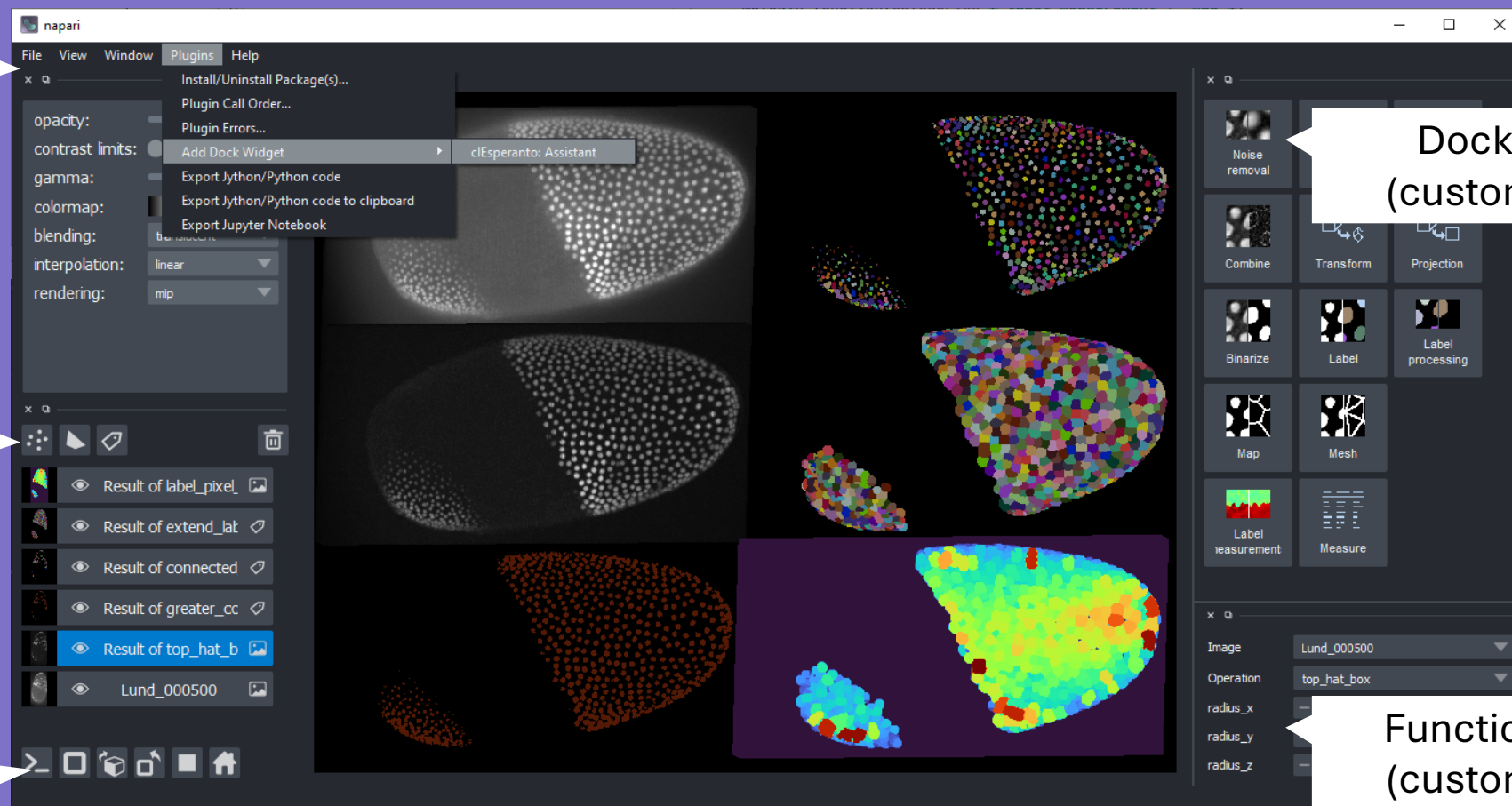
<https://napari.org/>

Interface napari: example ClEsperando plugin

View configuration /
tools

Layers

Viewer controls



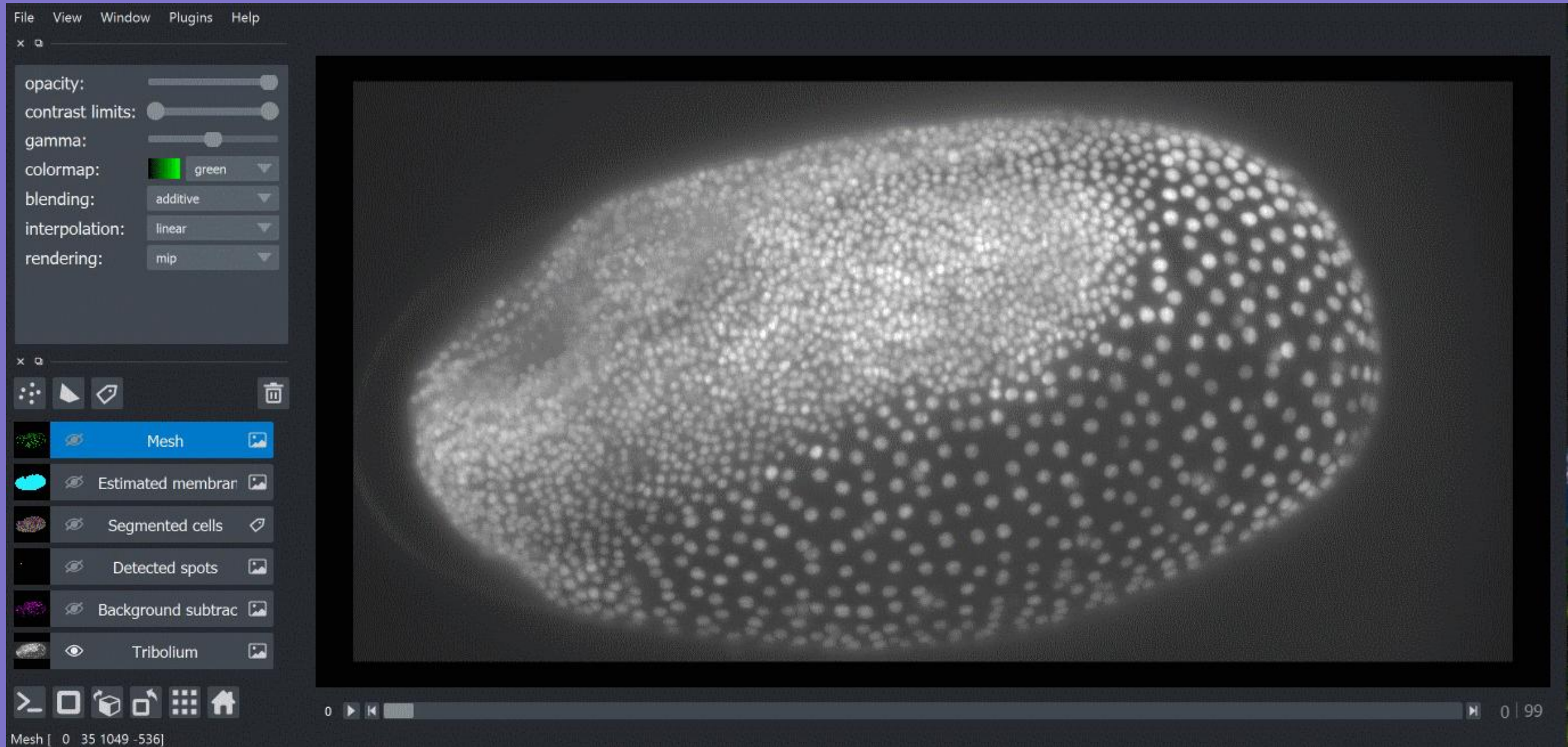
Dock widgets
(custom plugins)

Function widgets
(custom plugins)

Napari: 3D viewer for Python



Napari: 3D viewer for Python example with Mesh



Movies from Robert Haase : <https://haesleinhuepf.github.io/>

napari vs ImageJ

Tools Type

Python-based multi-dimensional viewer

Image processing tool for scientific research

Advantages

- Native 3D/4D visualization
- Modern, intuitive interface
- Built-in Python plugins
- Python ecosystem compatibility
- High performance
- Big data

- Easy to use
- Quick learning curve
- Built-in specialized tools
- Wide plugin support
- Quantitative analysis
- Supports medical formats

Tools Type

- Multi-dimensional visualization
- Interactive segmentation
- Volumetric image analysis
- Jupyter integration

- Measurements and annotations
- Filters and transformations
- Particle analysis
- Stacks and time series

Disadvantages

- Smaller user community (Actually)
- Fewer specialized plugins

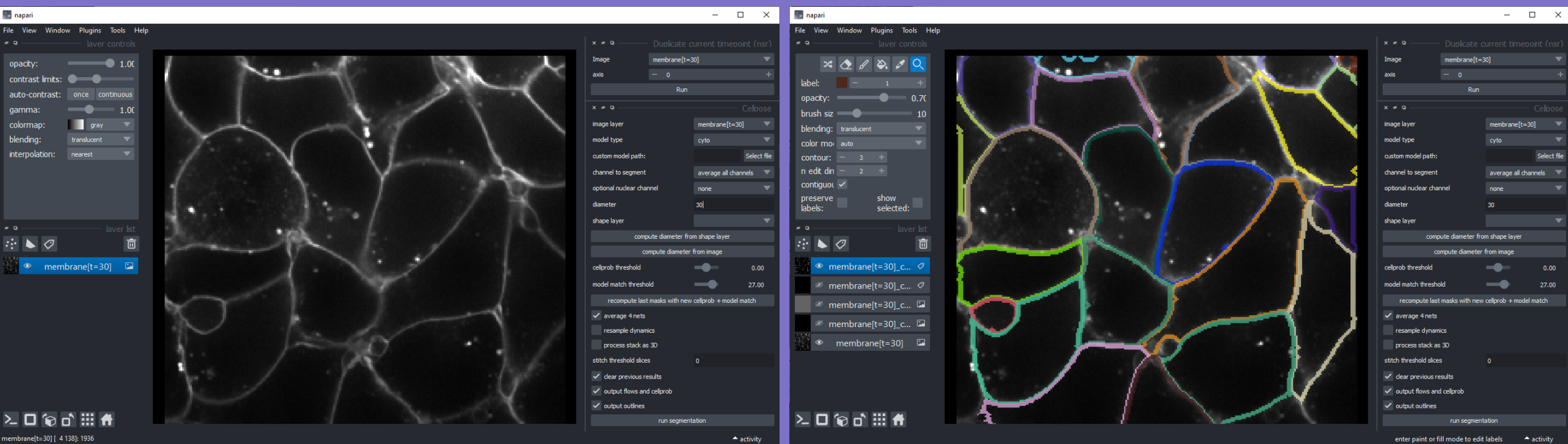
- Limited to predefined tasks
- Less flexible for automation
- Visualization in 3D
- Big data



Napari

Some Plugins

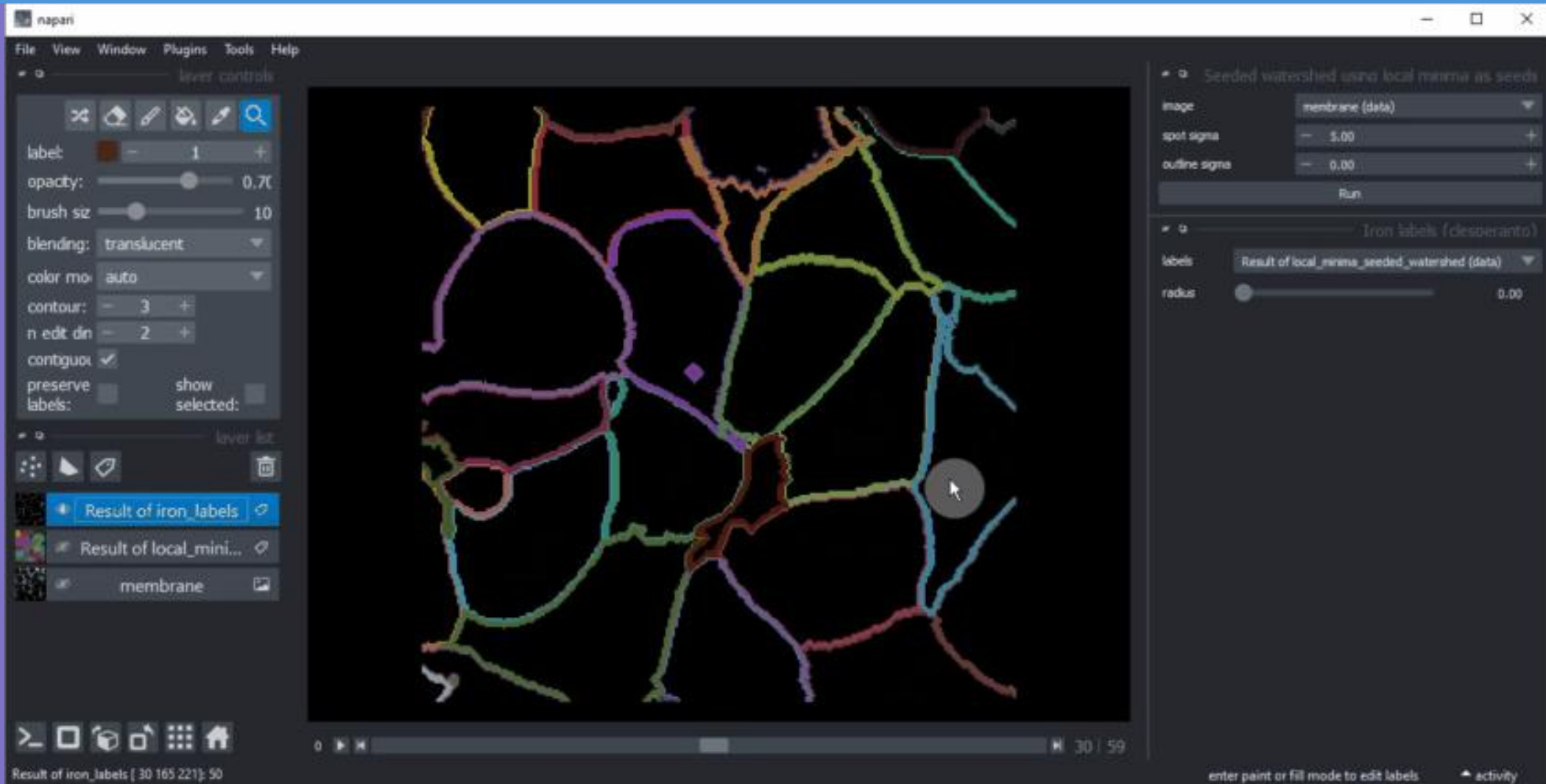
Deep learning based cell: Cellpose



<https://www.youtube.com/watch?v=bbcLt4BuMXo>

Stringer et al *Nat Methods* **18**, 100–106 (2021). <https://doi.org/10.1038/s41592-020-01018-x>
<https://cellpose-napari.readthedocs.io/en/latest/>

Napari: clEsperanto

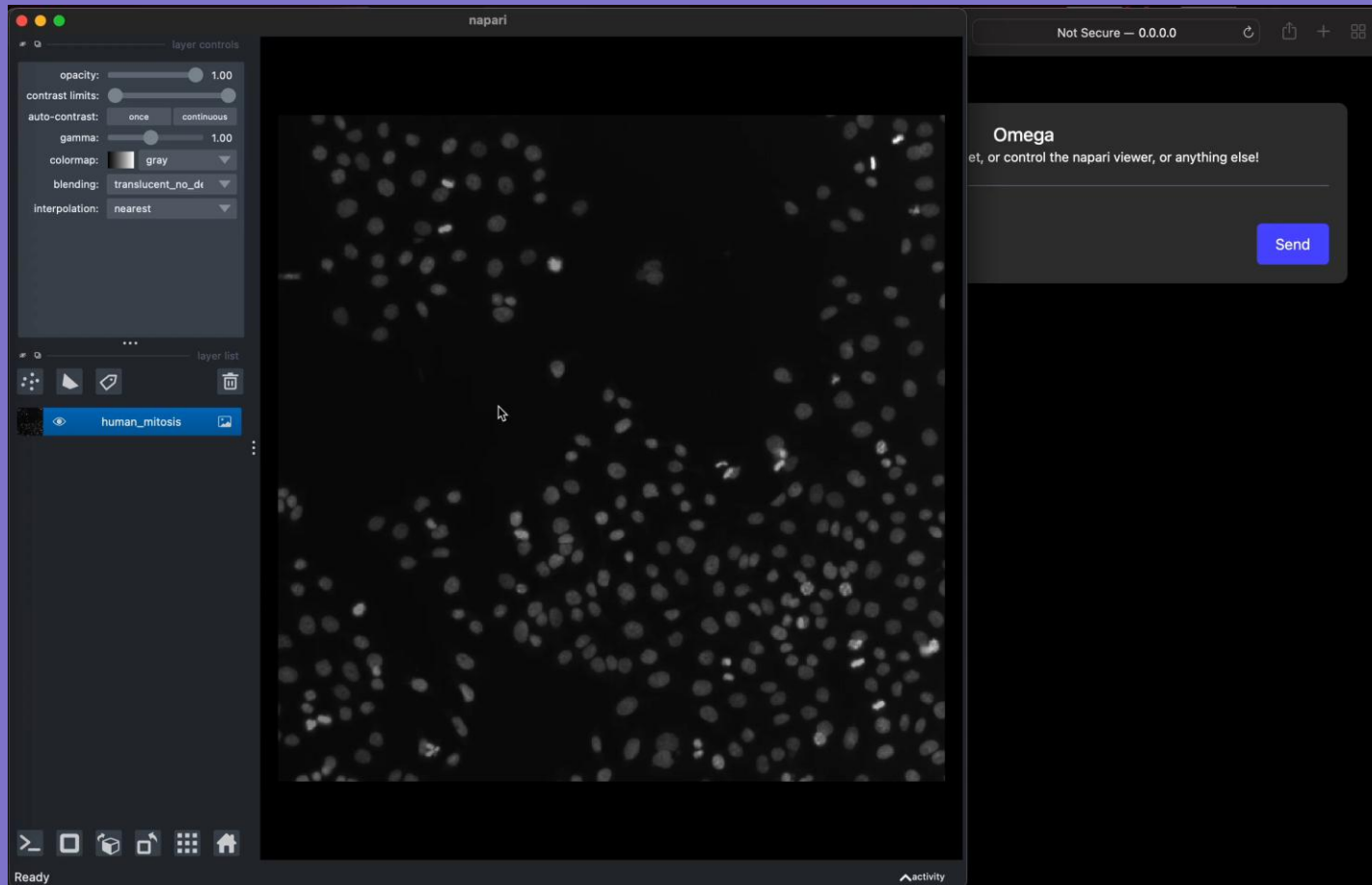


- In Napari menu Tools > Segmentation post-processing > Smooth labels (clEsperanto)

https://github.com/clEsperanto/napari_pyclesperanto_assistant

https://haesleinhuepf.github.io/BioImageAnalysisNotebooks/15_gpu_acceleration/clesperanto.html

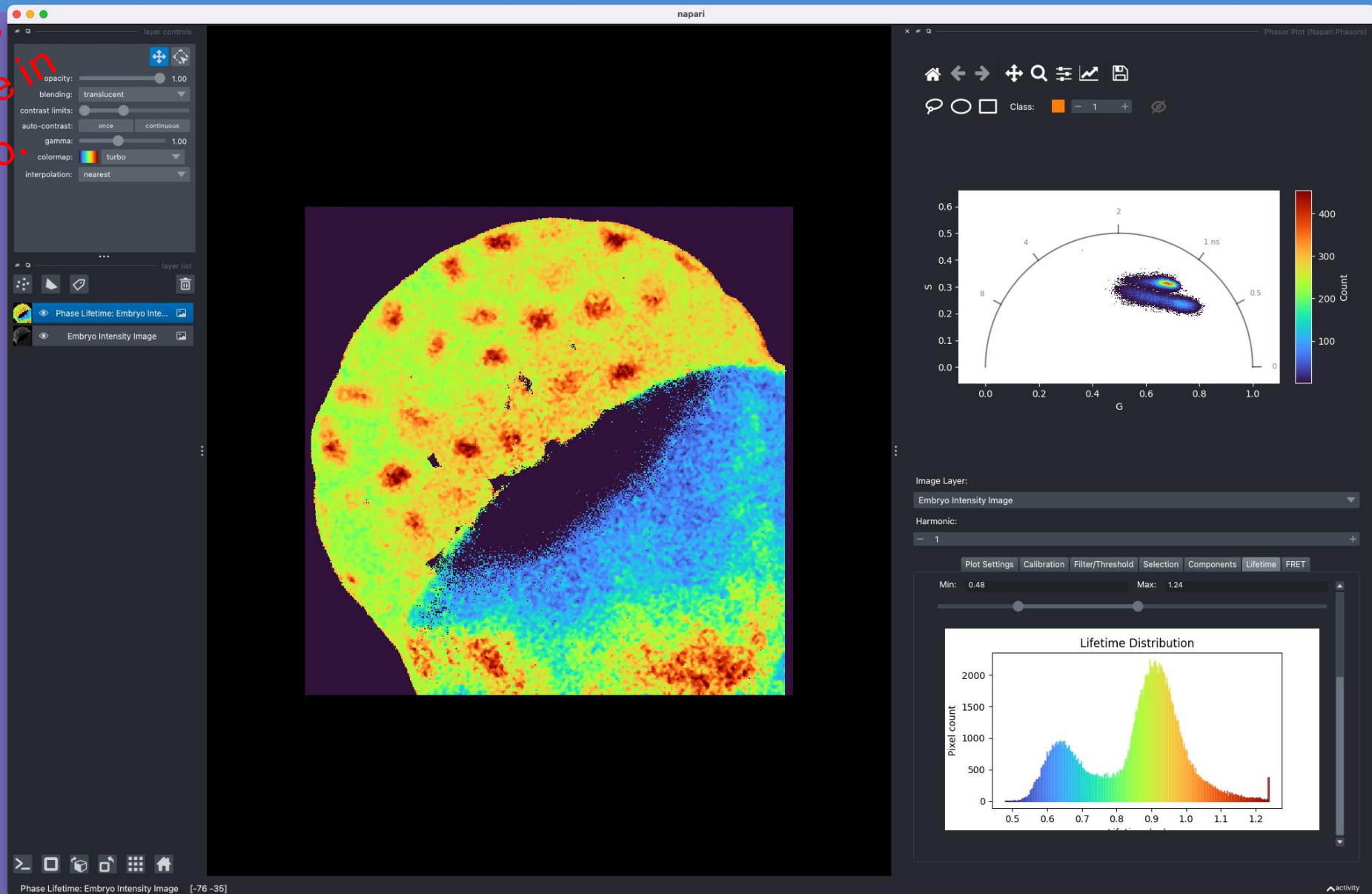
napari: chatGTP



Napari-chatGPT can automate programming plugins / “widgets”.

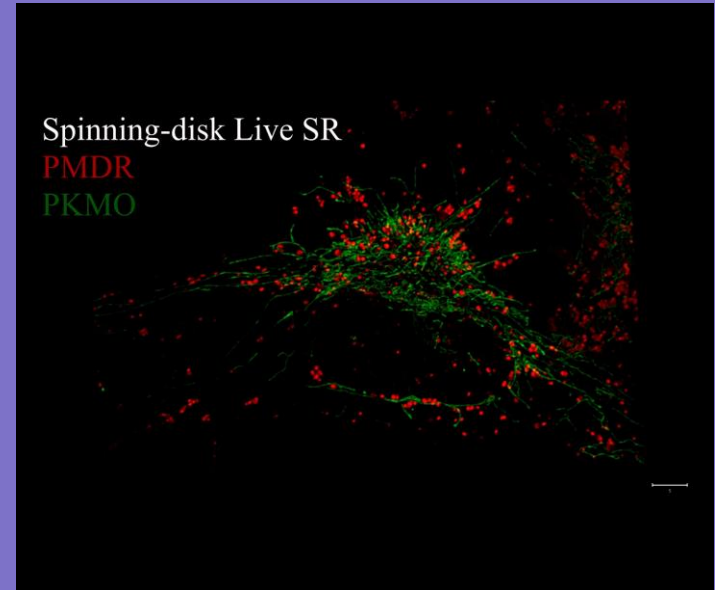
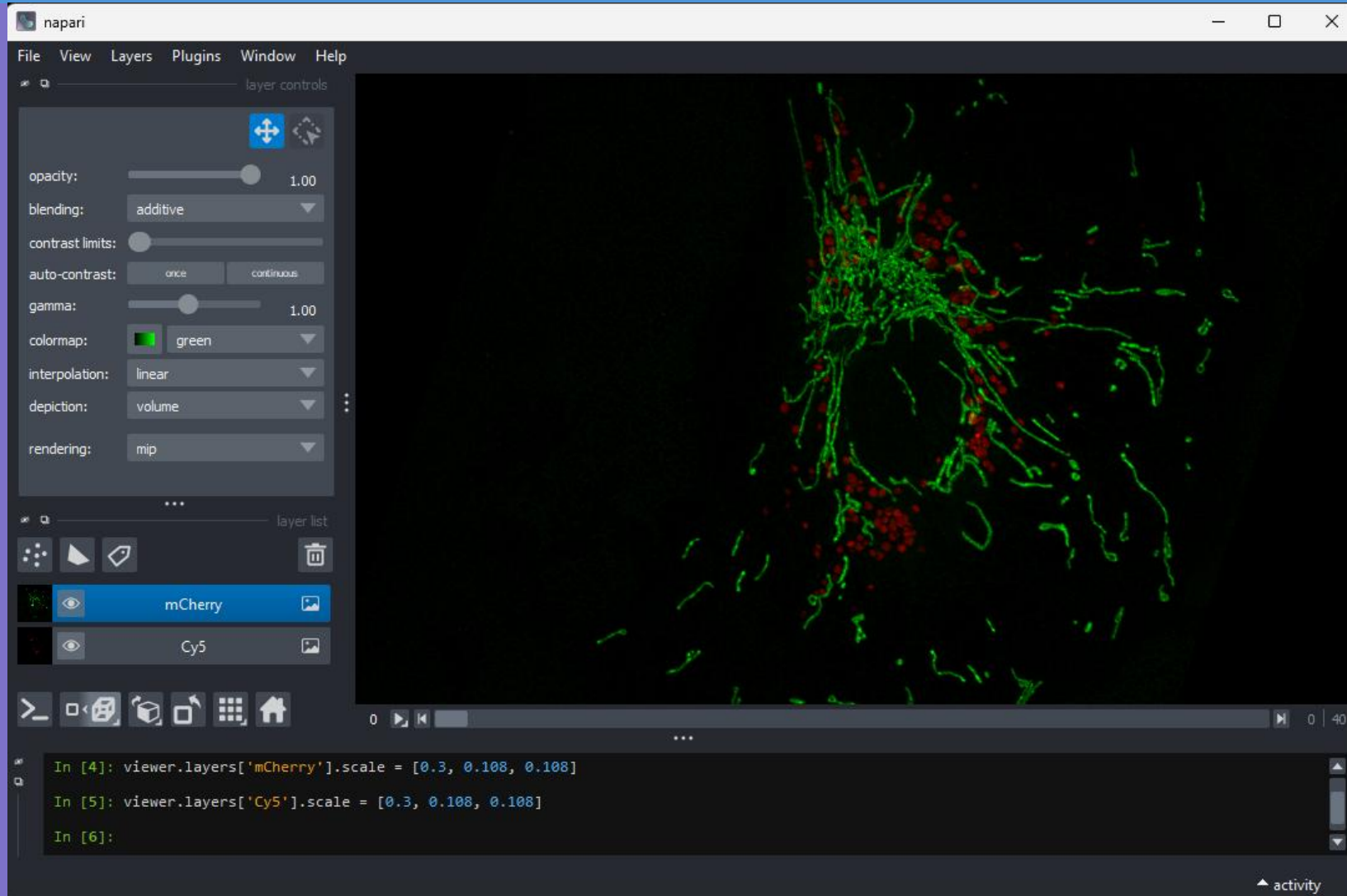
napari: PhasorPy

Produced at the
Pasteur Institute in
Montevideo.



Plugin by Bruno Pannunzio et al <https://github.com/napari-phasors/napari-phasors>
<https://napari-hub.org/plugins/napari-phasors.html>

napari: Animation



<https://www.blackmagicdesign.com/mx/event/davinciresolvedownload>

Plugin by Alister Burt, et al (BSD3 licensed) <https://github.com/napari/napari-animation>

Imaging data by B. Duygu Ozpolat et al (CC-BY 4.0 licensed) <https://doi.org/10.5281/zenodo.1063531>

napari: Installation

Manuel

<https://napari.org/0.6.1/tutorials/fundamentals/installation.html>

Simple Version, no necessary to install the all processing...
Installation plugins in napari' software

Automatic

<https://github.com/napari/napari/releases/tag/v0.6.2>

Step 1: Installing Pixi windows

1) Open PowerShell as Administrator

```
powershell -ExecutionPolicy ByPass -c "irm https://pixi.sh/install.ps1 | iex"
```

2) Restart PowerShell

3) Verifiy the installation

```
pixi --version
```

Step 2: Installing Napari

1) Creat project Napari

```
pixi init napari-project cd napari-project
```

2) Add napari and dependancies

```
pixi add python=3.11 napari pyqt scikit-image matplotlib  
pixi add python=3.11 napari  
pyqt scikit-image matplotlib
```

3) Launch napari

```
pixi run napari
```

Contributors



jni, aganders3, and 12 other contributors

Assets 12

napari-0.6.2-Linux-x86_64.lockfile.txt	sha256:80059e5f467eaf0936e2fbcc2329...		39.4 KB	last month
napari-0.6.2-Linux-x86_64.sh	sha256:ab520c613390a8421e1ba22645174...		461 MB	last month
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napari-0.6.2-Windows-x86_64.lockfile.txt	sha256:374266488efc4dec2f5c8f05c291e...		31.2 KB	last month
napari-0.6.2.tar.gz	sha256:c74c0fbcf087faa524475d5ef6d16...		3.05 MB	last month
Source code (zip)				last month
Source code (tar.gz)				last month

4 people reacted

napari: Installation with Pixi for Mac and Linux

Step 1: Install Pixi

```
# Install Pixi using the official script
curl -fsSL https://pixi.sh/install.sh | bash

# Reload your Shell
source ~/.bashrc # or ~/.zshrc
```

Step 2: Create a Napari Project

```
# Create project directory
mkdir my-napari-project
cd my-napari-project

# Initialize Pixi project
pixi init
```

Step 3: Add Napari & plugins

```
# Add Napari from conda-forge
pixi add napari

# Optional: Add useful plugins
pixi add napari-console
pixi add napari-svg
pixi add napari-plot
```

Step 4: Launch napari

```
# Run Napari in the project environment
pixi run napari

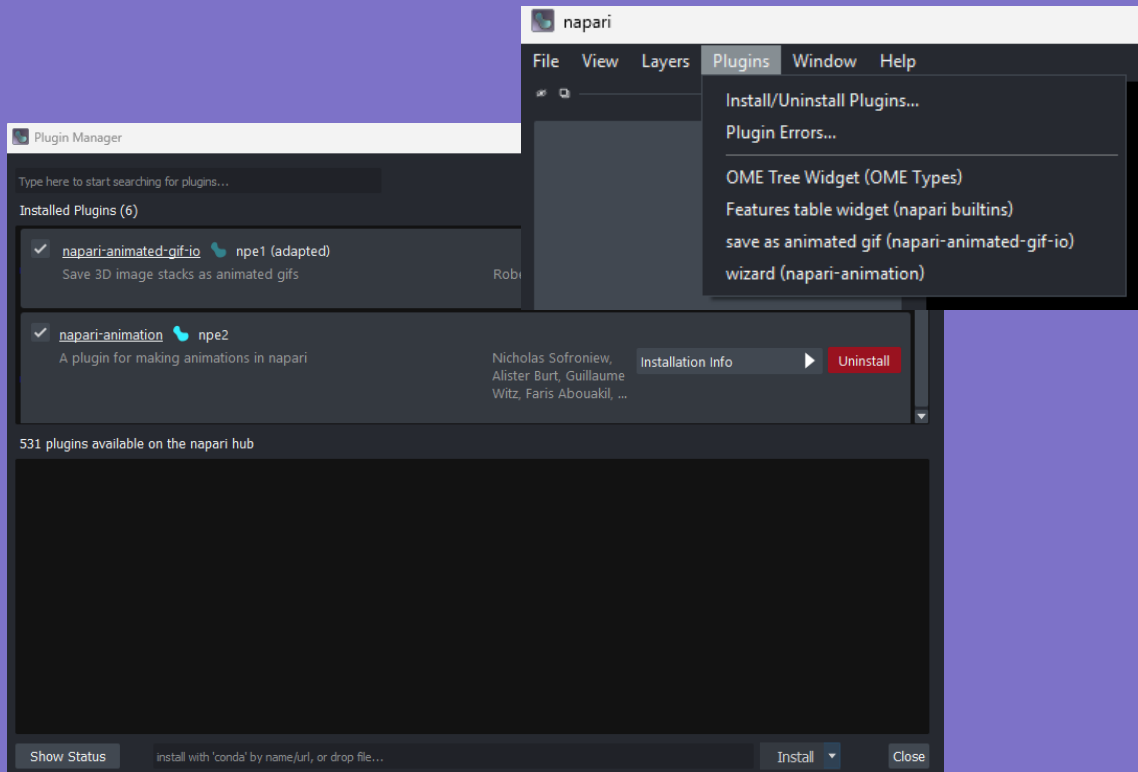
# Alternative: Global installation
pixi global install napari
```

napari: How to launch and install Plugins

Launch with PowerShell

```
PS C:\Users\Ludovic Leconte> cd napari-project
PS C:\Users\Ludovic Leconte\napari-project> pixi run napari
|
```

Install Plugins with two options



<https://napari.org/>

1) Close Napari

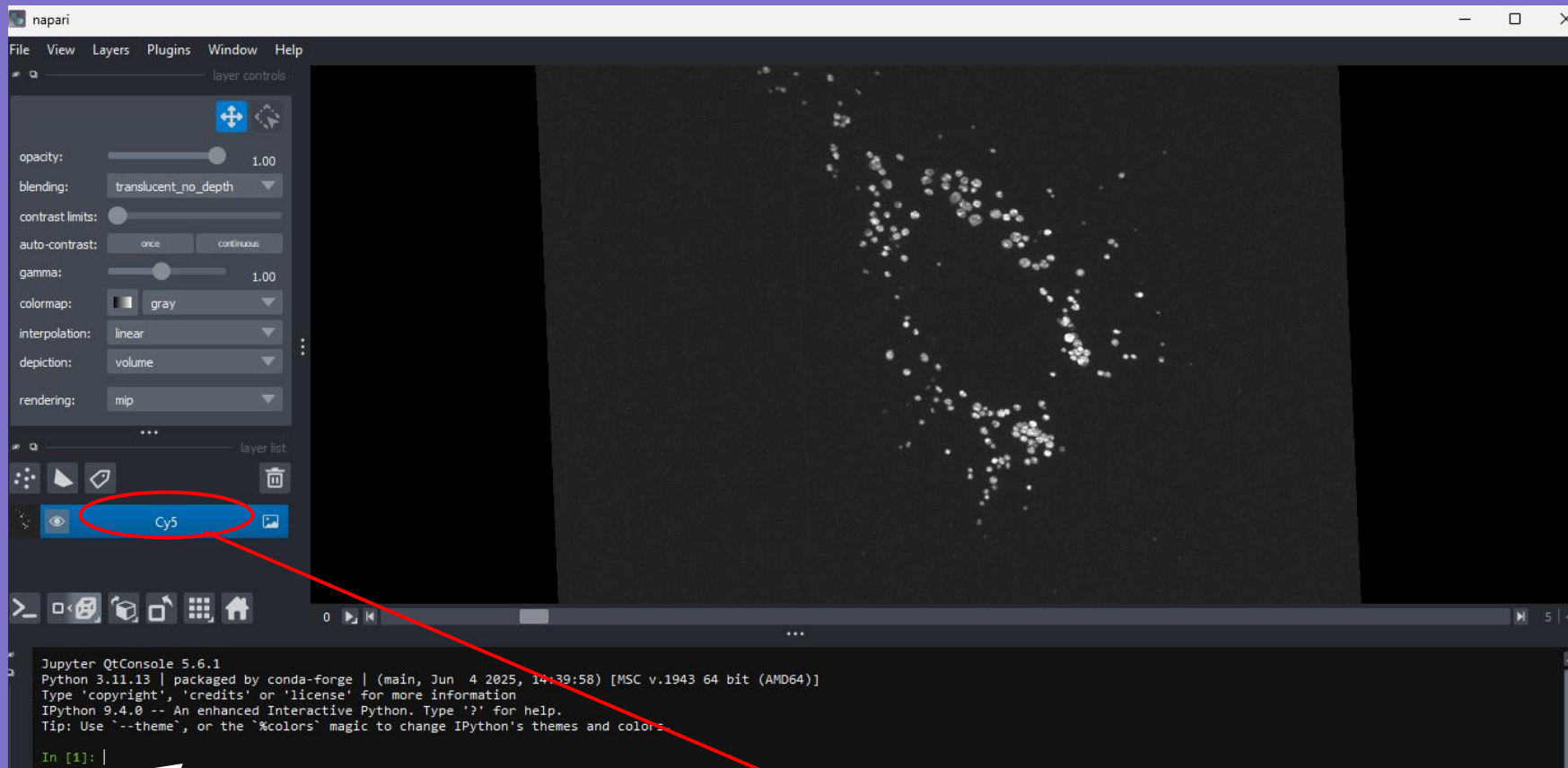
2) Powershell

C:\Users\Ludovic Leconte\napari-project

3) Verifiy the installation

Example: pip install napari-animation

napari: How to set the voxel size and contrast limits



Viewer controls

Voxel size

```
viewer.layers['Name'].scale = [z, x, y]
```

Contrast limits

```
viewer.layers['Name'].contrast_limits=(110,800)
```

napari: Some very important plugins

Bio-Format

AICSImageIO

<https://github.com/AllenCellModeling/aicsimageio>
pip install aicsimageio

ZARR (big Data)

napari-ome-zarr:

<https://github.com/ome/napari-ome-zarr>
pip install napari-ome-zarr

Create Videos

napari-animation:

<https://napari.org/napari-animation/>
pip install napari-animation

Segmentations

napari-sam

<https://github.com/MIC-DKFZ/napari-sampip>
install
napari-animation

napari-segment-blobs-and-things-with-membranes

<https://github.com/haesleinhuepf/napari-segment-blobs-and-things-with-membranes>
napari-animation

Analyse Particles

napari-skimage-regionprops

<https://github.com/haesleinhuepf/napari-skimage-regionprops>
pip install napari-skimage-regionprops

chatgtp

napari-chatgpt

<https://github.com/royerlab/napari-chatgpt>
pip install napari-skimage-regionprops